Graphs: Breadth-First-Search (BFS), Depth-First-Search (DFS), nents & topological numbering, Applications.

Trees: Creation & representation, Traversal, Copying, Printing an pretations of trees.

Memory Management: Uniform size records- explicit release and

Diverse Size Records: Allocation, Compaction.

Searching Techniques: Concept, Searching linked lists and Binary connected compo-

Hashing: Extraction, Compression, Division and Multiplication, C Chaining, Probing. d Arithmetic inter-

Collision Resolution, Double hash, ordered hash, Rehash, Radix d

Sorting: Discussion and comparison on different kinds of sorting garbage collection. Bubble sort, Quick sort, Selection sort, Merge sort etc.).

Recommended Texts: tree search.

1. E. M. Reingold and W. J. Hansen, *Data Structures*. Little, Br ollision Resolution:
2. S. Lipschutz, *Schaum's Outline of Theory and Problems* McGraw-Hill, 1986 istribution.

(i.e. Insertion sort,

CSE 4305 Computer Organization and Architecture

Components of a computer system: processors, memory, seconda and media, and other input output devices.

Processor organization: registers buses multiplexers decoders own, 1983

of Data Structures.

Credit: 3.00

ry storage devices

, , , , ALUs, clocks, main memory and caches.

Information representation and transfer; instruction and data access methods;

The control unit: hardwired and microprogrammed; memory organization, I/O systems, channels, interrupts, DMA. Von Neumann SISD organization. RISC and CISC machines.

Recommended Texts:

1. J. P. Hayes, *Computer Architecture and Organization*, 3rd ed. William C Brown Pub, 1997

CSE 4307 Database Management Systems Credit: 3.00

)verview of database management systems; DBMS file structures; introduction to he relational model; relational algebra, normalization and relational design; ER nodeling, object-oriented modeling, advanced features of the relational model; Data- ase Design Language; the hierarchical model; the CODASYL model; alternative lata models; physical database design; fourth-generation environment; database dministration, database recovery, distributed databases and current trends in the eld. Relational query languages: SQL; embedded SQL in a third-generation language COBOL, C or C++). Transaction management; concurrency control.

Recommended Texts:

1. A. Silberschatz et al., *Database System Concepts*, 7th ed. McGraw-Hill, 2019
2. C. J. Date, *An Introduction to Database Systems*, 8th ed. Pearson, 2003

CSE 4309 Theory of Computing Credit: 3.00

Review of Discrete Mathematics - Binary relations, digraph, string, languages, proofs, nductive definitions.

Formal methods of automata language and computability, Finite automata and regular expressions, Properties of regular sets, Contextfree grammars, Push-down automata, Properties of context-free languages, Turing machines, Halting problem, Undecidabil- ity and Computability, Recursion function theory, Chomsky hierarchy, Deterministic context-free languages, Closure properties of families of languages, Computational complexity theory, Intractable problems, Applications in parsing, pattern matching and the design of efficient algorithms.

Finite state machines, Introduction to sequential circuits, basic definition of finite state model, memory elements and their excitation functions, synthesis of synchronous sequential circuits, iterative networks, definition and realization of Moore and Mealey machines.

Recommended Texts:

1. J. E. Hopcroft et al., *Introduction to Automata Theory, Languages, and Computation*, 3rd ed. Pearson, 2006
2. M. Sipser, *Introduction to the Theory of Computation*, 3rd ed. Cengage Learning, 2012

SWE 4301 Object Oriented Concepts II Credit: 3.00

Review of Object Oriented Concept, Multi-threading, UML Diagrams for Class, Objects and Relationships, UI programming, Synchronizations, Client Server programming, RPC, Distributed Objects, XML, Web programming : URL, Request and Response,